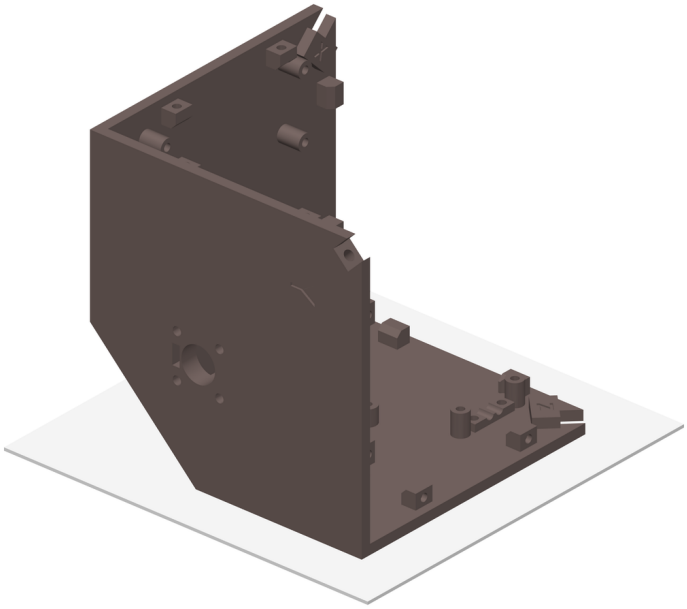
Printing

Every printed part, in the order you need it



Designed by Athanasia Nikolova

v1.1 · October 2026

All data come from tests and many real builds, not only from theory.

Teacher only

- **One thing per page**
  One action, one big picture. The part of the step is the only strong colour, the rest is quiet grey. Short sentences, much white space.
- **Colour is never the only clue**
  Every wire on a board drawing also has a number, every board a letter (X, Y, Z), every insert its size. The insert colours stay distinct with colour-vision deficiency (Okabe-Ito palette).
- **Readable**
  All instruction text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).
- **Print it right**
  In colour, at 100 % - not 'fit to page'. The Assembly guide draws every screw at real size: the student lays the screw on the page to check it.
- **Pace**
  One page, then the action, then the next page. For a student who tires quickly: one chapter per session. The check pages are good stopping points.

✋ Safety first

- Work with your teacher or mentor in the room.
- The nozzle and the bed are HOT. Never touch them while the printer works or just after.
- Hands out of the printer while it moves.
- Open a window: the warm plastic smells.
- Take a part off only when the bed is cool. Push the scraper AWAY from your hand.
- Carbon-fibre filament: no sanding without a mask - its dust is not for breathing.

Before you start

- The filament is PETG-CF (PETG with carbon fibre). It wears a soft brass nozzle - use a hardened steel nozzle.
- Keep the filament dry: closed bag with a drying pack, or dry it before printing.
- Two colours: Milk Tea Brown and Ancient Wood Green.
- The files in CAD/print are already turned the right way: put them on the bed as they are.
- Your printer's bed: at least 166 x 123 mm, and 131 mm high (the longest part and the box).

The three settings

Every part uses one of them - its page says which.

Standard - most parts

- Layer height 0.16 mm · 4 walls (wall loops)
- Top: 5 layers (0.8 mm) · bottom: 4 layers
- Infill 20 %, rectilinear pattern · no support

Fine - small and precise parts

- Layer height 0.08 mm · 5 walls (wall loops)
- Infill 20 %, rectilinear pattern · no support (the navel: WITH support)

Heavy - the reaction wheels

- Layer height 0.16 mm · 4 walls (wall loops)
- Infill 90 %, rectilinear pattern · no support - the wheels must be heavy

All three

- Speeds: first layer 50 mm/s · inner wall 220 mm/s · outer wall 170 mm/s
- PETG-CF: nozzle 255 °C · bed 75 °C · fan 40 %
- Dry the filament 8 hours at 65 °C, and print straight from the dryer.

How to read the pages

- The picture: the part as it lies on the bed.
- Top right: how many you print.
- The colour dot: which filament.
- Settings: layer height, walls, infill, support - set them in your slicer.
- Grams: from the 3D model (or weighed, where it says so). Good for planning the filament.

All the parts

In the order the build needs them.

1 The box and the wheels

Box	× 1	172 g
Reaction wheel	× 3	282 g
Magnet carrier	× 3	2 g

2 Battery holder and IMU

Battery tube	× 1	22 g
Battery basin	× 1	30 g

3 Wiring

clamp_power	× 1	0 g
clamp_z	× 1	2 g
clamp_y	× 1	3 g
clamp_x_1	× 1	1 g
clamp_x_2	× 1	1 g
Clamp spacer	× 8	1 g

4 The cage

Box standoff	× 1	6 g
Hub standoff	× 1	3 g
Cage corner	× 8	21 g
Cage beam	× 9	83 g
Tie beam	× 3	27 g

5 Covers

Cover	× 3	58 g

6 The sphere

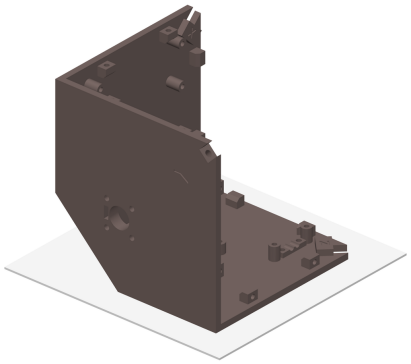
Navel	× 8	66 g
Edge arc	× 12	90 g
Diagonal arc, lower	× 6	77 g
Diagonal arc, upper	× 6	77 g

Ancient Wood Green	**about 467 g**
Milk Tea Brown	**about 556 g**

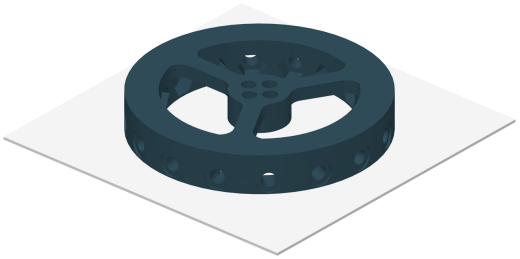
Plus about 10 % for the first layers, supports and a failed print.

1 · The box and the wheels

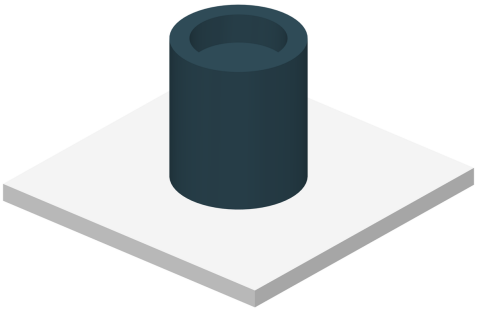
Chapter 1 of the Assembly guide.



Box × 1



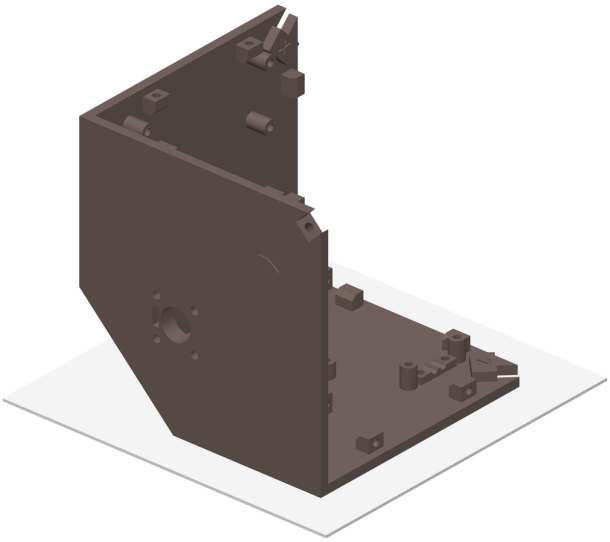
Reaction wheel × 3



Magnet carrier × 3

1 Box

× 1



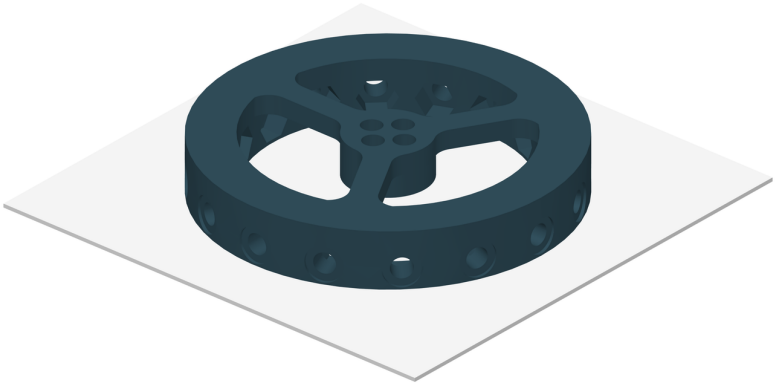
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 172 g each.
- The biggest print: it needs the whole height of the printer.
- It was printed and weighed: 172 g with these settings.

File: CAD/print/box.step

2 Reaction wheel

× 3



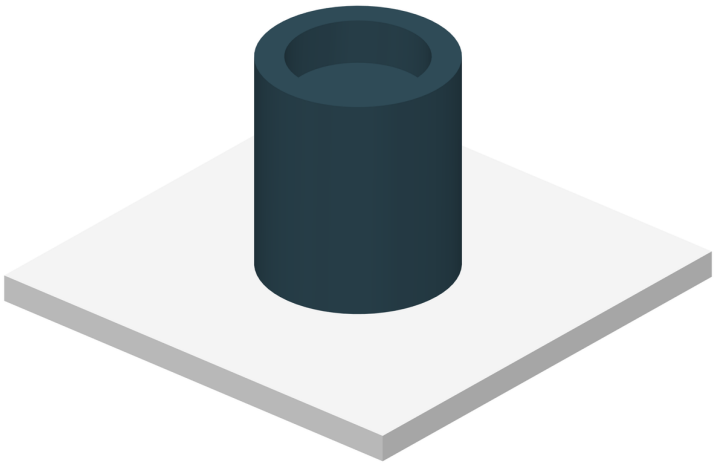
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 90% infill · no support - the heavy settings
- About 94 g each · 282 g for all 3.
- Flat on the bed, as in the picture.
- Heavy settings (90 % infill): its weight is what balances the cube.

File: CAD/print/reaction_wheel.step

3 Magnet carrier

× 3



Ancient Wood Green · PETG-CF

- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 1 g each · 2 g for all 3.
- Fine settings: it holds the magnet exactly in the middle.

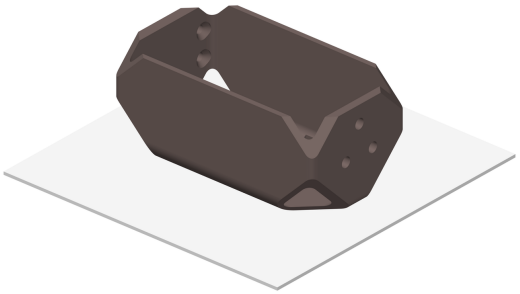
File: CAD/print/magnet_carrier.step

2 · Battery holder and IMU

Chapter 2.



Battery tube × 1



Battery basin × 1

1 Battery tube

× 1



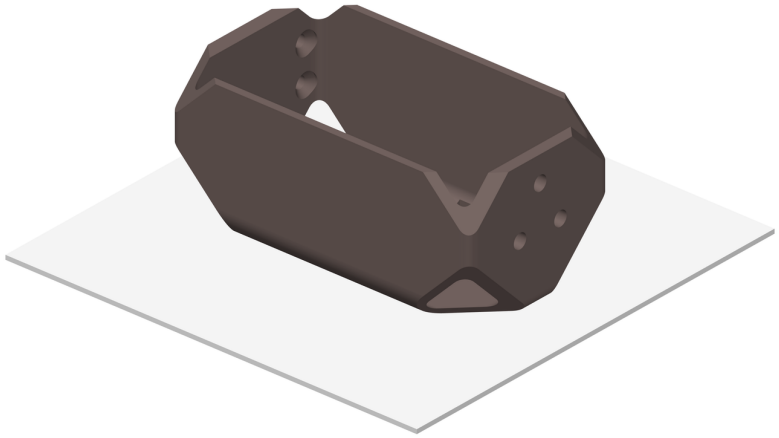
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 100% infill · no support
- About 22 g each.
- SOLID (100 % infill): the IMU sits in its table - it must not bend or shake.
- The foot on the bed.

File: CAD/print/battery_tube.step

2 Battery basin

× 1



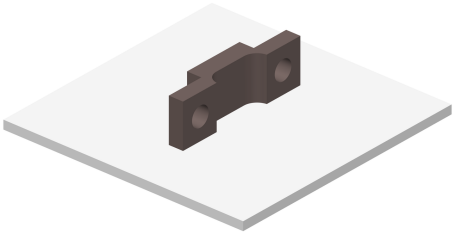
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 100% infill · no support
- About 30 g each.
- Print it ON ITS BACK, as in the picture.

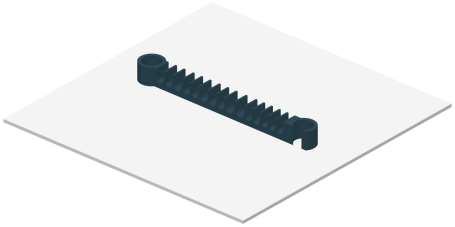
File: CAD/print/battery_basin.step

3 · Wiring

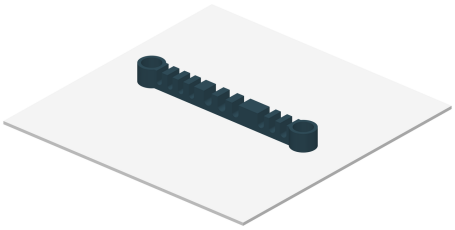
Chapter 3 - clamp_power, the 4 cable clamps and their spacers.



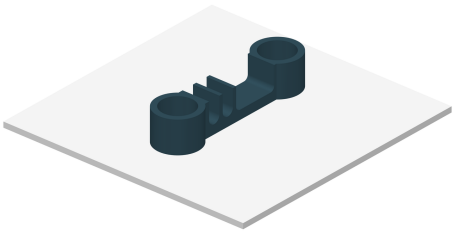
clamp_power × 1



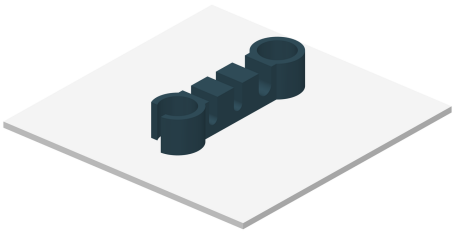
clamp_z × 1



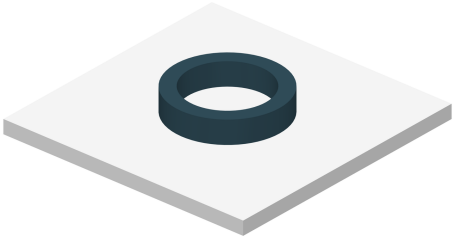
clamp_y × 1



clamp_x_1 × 1



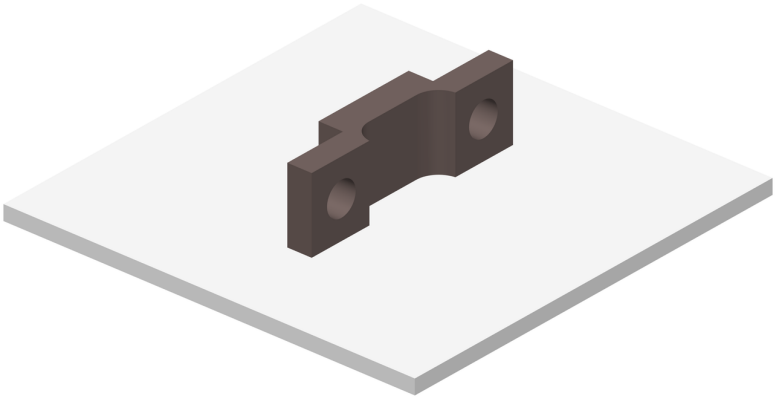
clamp_x_2 × 1



Clamp spacer × 8

1 clamp_power

× 1



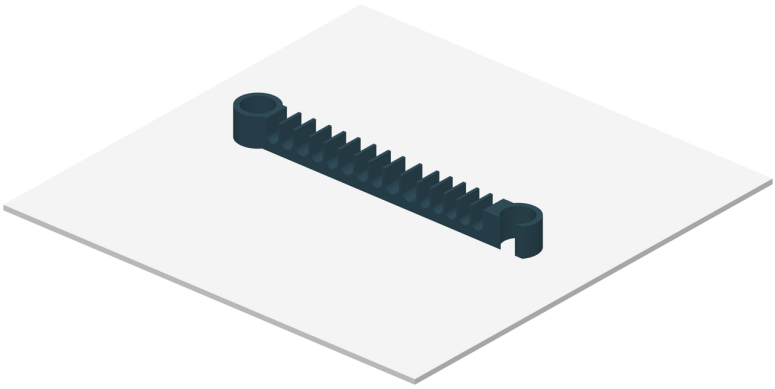
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 0 g each.

File: CAD/print/clamp_power.step

2 clamp_z

× 1



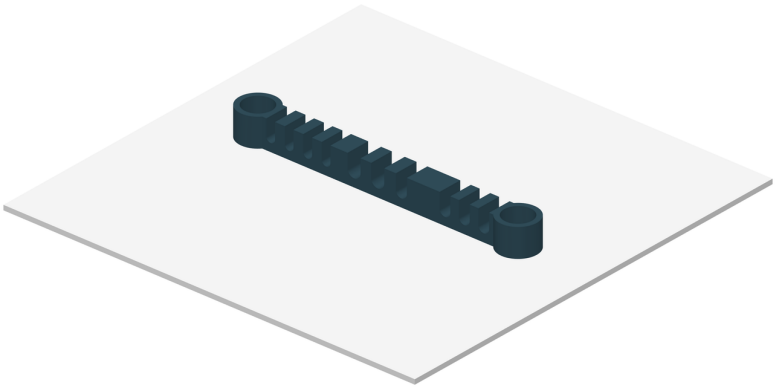
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 2 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_z.step

3 clamp_y

× 1



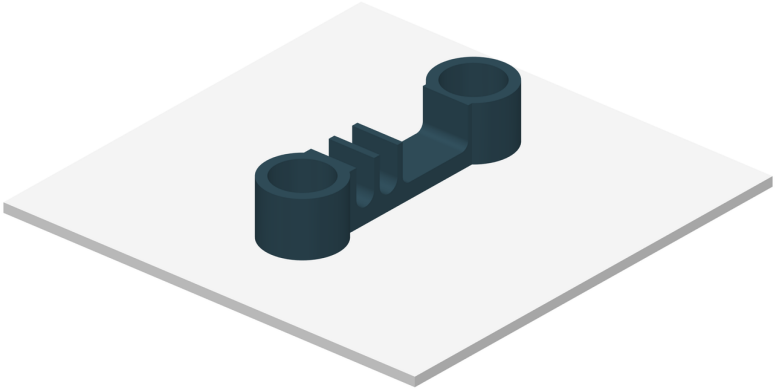
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 3 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_y.step

4 clamp_x_1

× 1



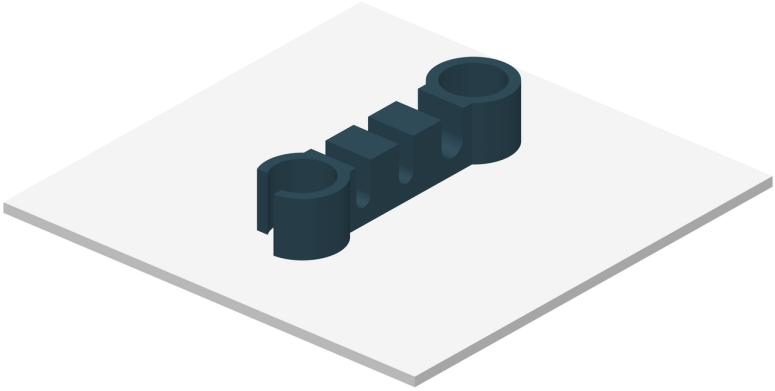
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 1 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_x_1.step

5 clamp_x_2

× 1



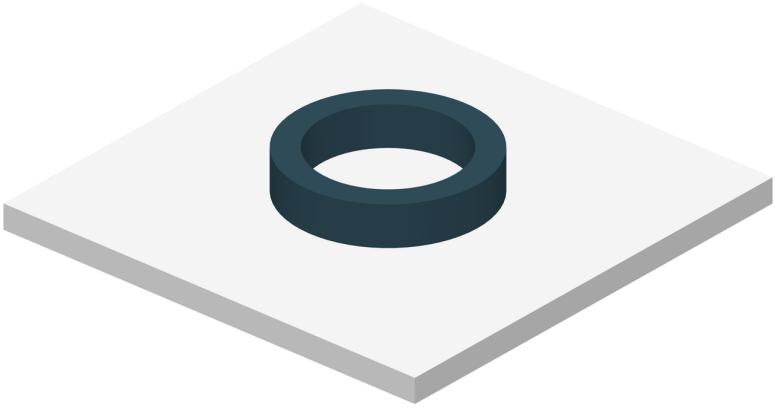
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 1 g each.
- Upside down, as in the picture: the flat top on the bed, the openings up.
- No support: the openings are open channels.

File: CAD/print/clamp_x_2.step

6 Clamp spacer

× 8



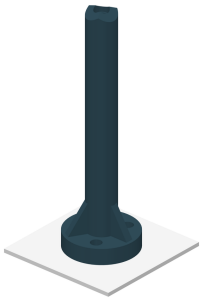
Ancient Wood Green · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 0 g each · 1 g for all 8.
- 8 of them: one on each clamp ring, under the board.
- Small: print them together with a bigger part.

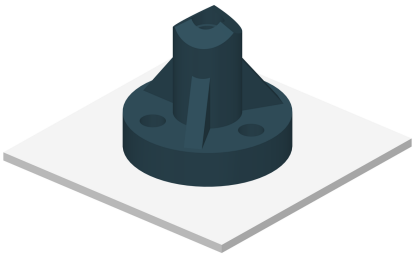
File: CAD/print/spacer_OD9.5_ID7_H2.step

4 · The cage

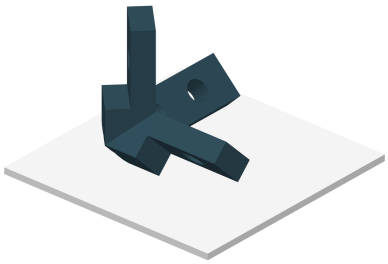
Chapter 4.



Box standoff × 1



Hub standoff × 1



Cage corner × 8



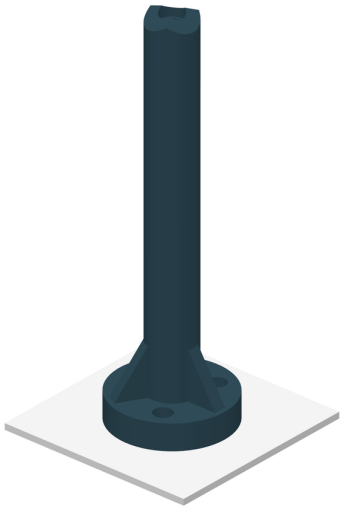
Cage beam × 9



Tie beam × 3

1 Box standoff

× 1



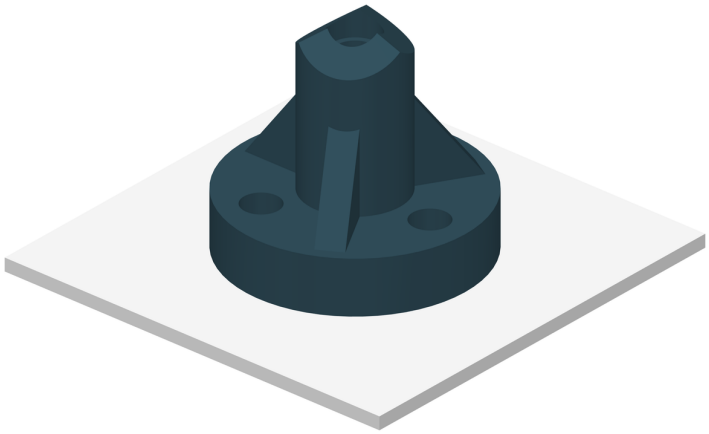
Ancient Wood Green · PETG-CF

- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 6 g each.

File: CAD/print/box_standoff.step

2 Hub standoff

× 1



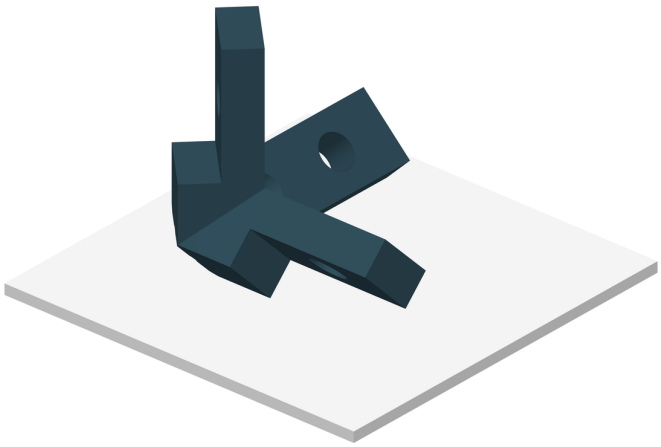
Ancient Wood Green • PETG-CF

- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 3 g each.

File: CAD/print/battery_top_standoff.step

3 Cage corner

× 8



Ancient Wood Green · PETG-CF

- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 3 g each · 21 g for all 8.

File: CAD/print/corner.step

4 Cage beam

× 9



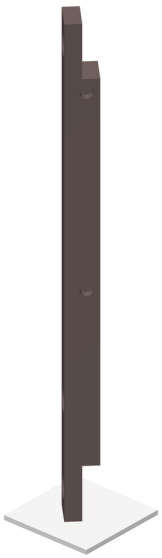
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 9 g each · 83 g for all 9.
- 9 plain beams. The other 3 beams are tie beams - the next part.

File: CAD/print/beam.step

5 Tie beam

× 3



Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 9 g each · 27 g for all 3.
- 3 tie beams: a hole for the box screw and 2 cover magnet pockets.

File: CAD/print/beam_tie.step

5 · Covers

Chapter 5.



Cover × 3

1 Cover

× 3



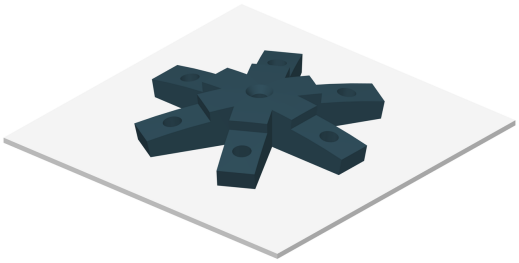
Ancient Wood Green · PETG-CF

- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- About 19 g each · 58 g for all 3.
- Fine settings: flat and exact, the magnets sit in it.
- Let it cool on the bed to room temperature before you take it off - then it stays flat.

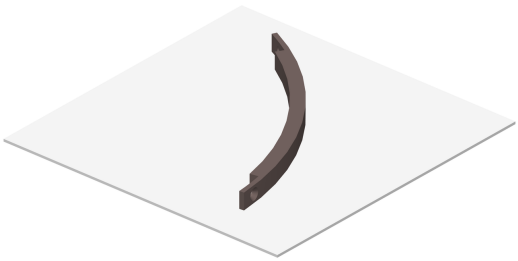
File: CAD/print/cover.step

6 · The sphere

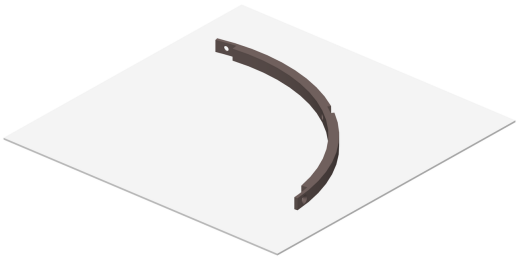
Chapter 6.



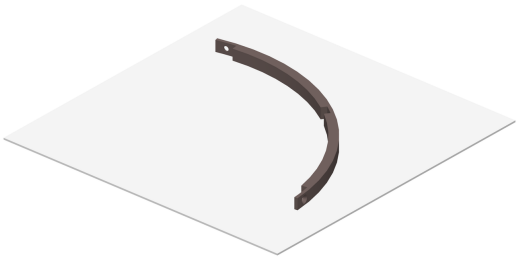
Navel × 8



Edge arc × 12



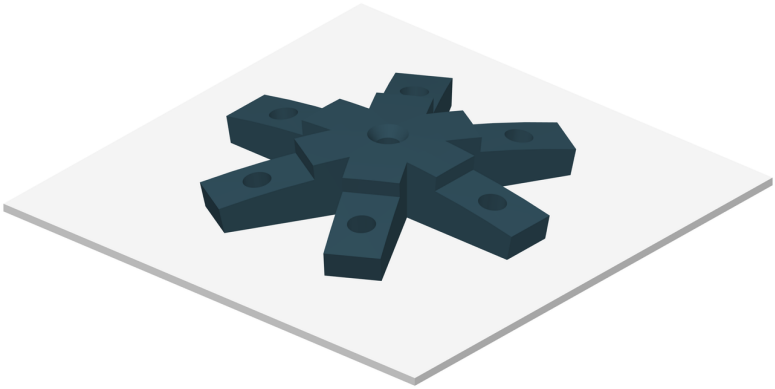
Diagonal arc, lower × 6



Diagonal arc, upper × 6

1 Navel

× 8



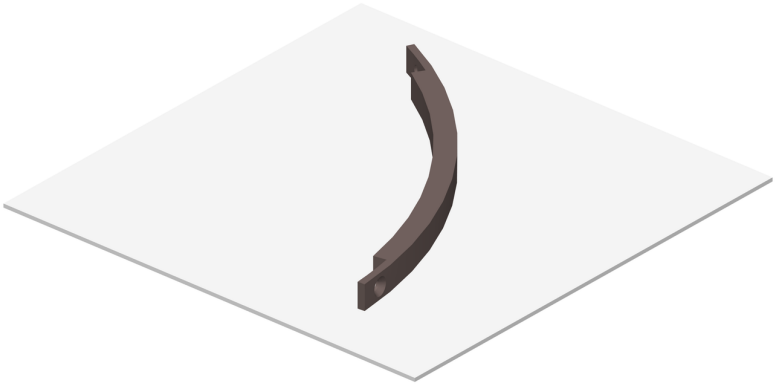
Ancient Wood Green · PETG-CF

- Layer 0.08 mm · 5 walls · 20% infill · WITH support - the fine settings
- About 8 g each · 66 g for all 8.
- WITH support - the only part that needs it.

File: CAD/print/navel.step

2 Edge arc

× 12



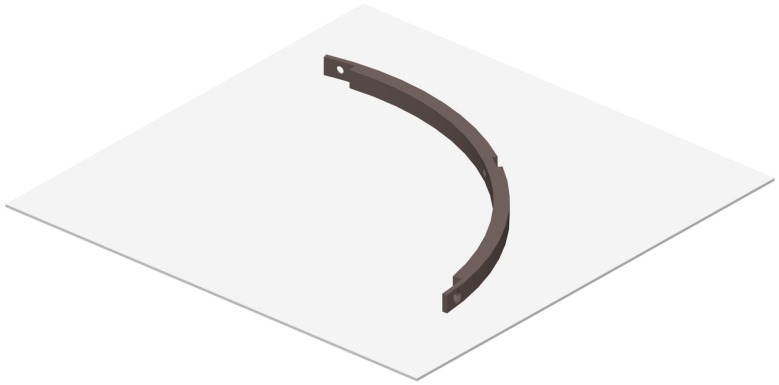
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 8 g each · 90 g for all 12.

File: CAD/print/sphere_arc_edge.step

3 Diagonal arc, lower

× 6



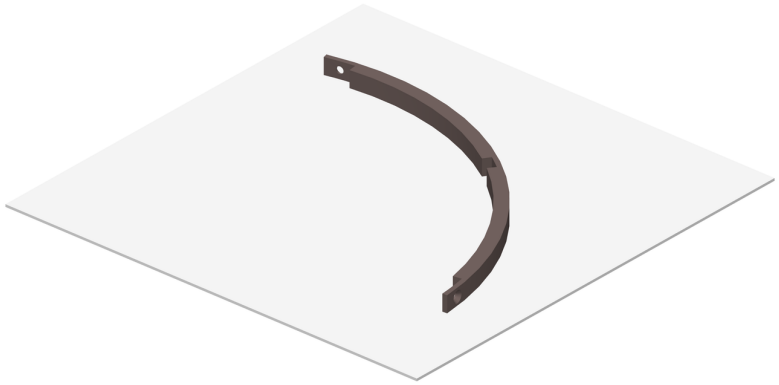
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 13 g each · 77 g for all 6.
- 6 LOWER diagonal arcs - they have the insert in the middle.

File: CAD/print/sphere_arc_diag_under.step

4 Diagonal arc, upper

× 6



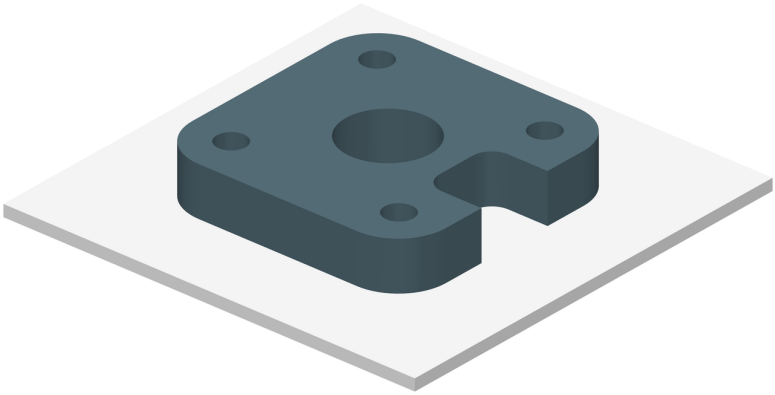
Milk Tea Brown · PETG-CF

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- About 13 g each · 77 g for all 6.
- 6 UPPER diagonal arcs.

File: CAD/print/sphere_arc_diag_over.step

Magnet-handler (tool)

× 1

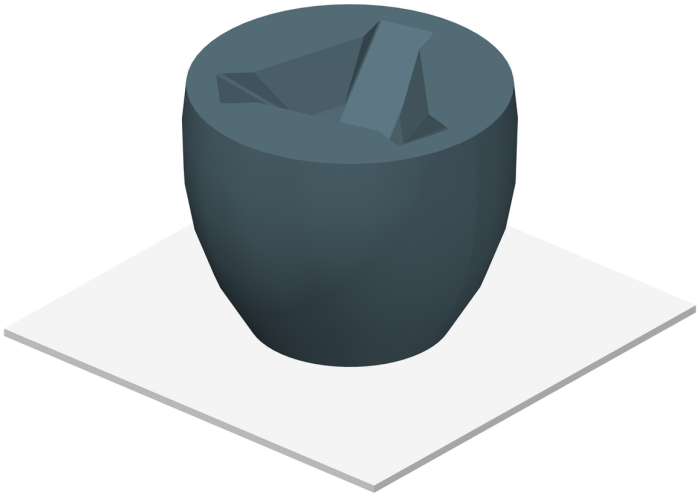


any colour · PETG-CF or PLA

- Layer 0.08 mm · 5 walls · 20% infill · no support - the fine settings
- Magnets guide: centres the magnet carrier.
- A tool: it is used, then put away. Any colour.

File: CAD/print/magnet-handler.step

Display base, top (not part of the cube) × 1

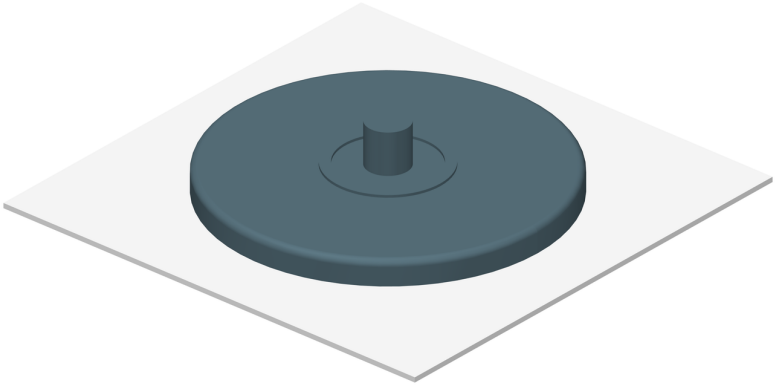


any colour · PETG-CF or PLA

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- the cube stands on corner 0 in its cup.
- Display base, top: hollow inside, closed. Flat side on the bed.

File: CAD/print/base_top.step

Display base, bottom (not part of the cube) × 1

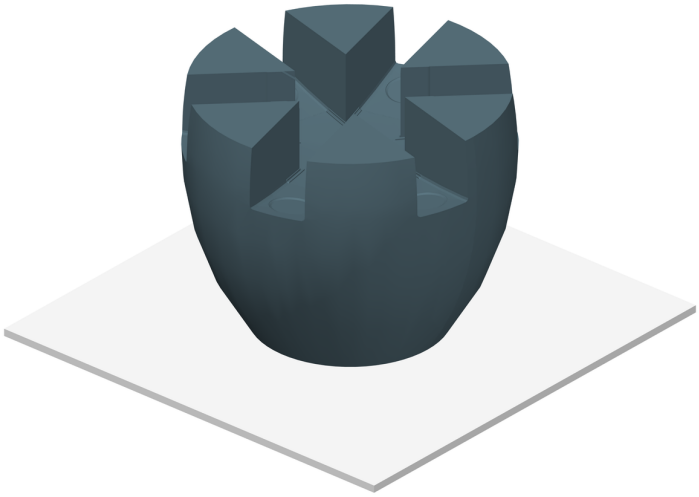


any colour · PETG-CF or PLA

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- the disc; the top turns on its pin.
- Display base, bottom: the disc. Flat side on the bed, the pin up.

File: CAD/print/base_bottom.step

Display base, top for the cube WITH the sphere (not pa



any colour · PETG-CF or PLA

- Layer 0.16 mm · 4 walls · 20% infill · no support - the standard settings
- the navel and its 6 arcs sit in it; the same bottom.
- Display base, top for the cube with the sphere: hollow inside, closed.
- Flat side on the bed. Print base_top OR this one - the bottom is the same.

File: CAD/print/base_top_sphere.step

After printing

- Wait until the bed is cool, then take the part off.
- Supports off (navels only) - with pliers, away from your hand.
- Look at every hole: no strings, no blobs. A drill bit by hand cleans a hole.
- Write the colour and the count on a list - tick each part.
- Next: the Inserts guide - the brass inserts go into the printed parts.